STRÜVER Model — Formula-Only (v4)

TRIAS Architecture, 70/30 Internal Rule, Dynamics, Hazard, Phase Regimes

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1 Index Set and Base Definitions

$$i, j \in \{X, Y, Z\}$$

$$u_i \geq 0$$

$$S = \sum_i u_i$$

$$\rho_i = \frac{u_i}{S}, \quad \sum_i \rho_i = 1$$

The state space of ρ lies on the 2-simplex.

Smooth helper functions:

$$\mathrm{sp}(x; \tau) = \tau \ln\!\left(1 + e^{x/\tau}\right)$$

$$\sigma(x; \tau) = \frac{1}{1 + e^{-x/\tau}}$$

2 Parameter Domains

$$0 < \alpha < \beta < 1$$

$$\theta \in (\beta, 1)$$

$$\tau_{\text{out}}, \tau_{\text{in}}, \tau_u, \tau_p > 0$$

$$k_1, k_2, k_3, k_4 \geq 0$$

$$a_i \in \mathbb{R}$$

$$b_i, c_i, \lambda_i, \chi_i \geq 0$$

$$\eta_{ij}, \zeta_i \in \mathbb{R}$$

3 Internal Structure per Pillar (70/30 Rule)

Each pillar decomposes into ordered and free components:

$$u_i = u_i^{\text{ord}} + u_i^{\text{free}}$$

$$p_i = \frac{u_i^{\text{ord}}}{u_i} \in [0, 1]$$

Target internal proportion:

$$p^\star = 0.30$$

Tolerance band:

$$\delta \in (0, 0.5)$$

Smooth internal deviation function:

$$d_i(p_i) = \text{sp}(p_i - (p^\star + \delta); \tau_{\text{in}}) + \text{sp}((p^\star - \delta) - p_i; \tau_{\text{in}})$$

4 Core Dynamic Equation

4.1 Primary u-Dynamics

$$\dot{u}_i = \Big(a_i + \sum_{j \neq i} \eta_{ij} \rho_j\Big) u_i - \Big(b_i \rho_i + c_i \rho_i^2 + \lambda_i d_i(p_i)\Big) u_i + \zeta_i \tag{1}$$

This defines growth, cross-interaction, nonlinear saturation, and internal structural correction.

5 Derived Quantities

$$\dot{S} = \sum_i \dot{u}_i$$

$$\dot{\rho}_i = \frac{\dot{u}_i S - u_i \dot{S}}{S^2}$$

6 Diagnostics

6.1 Balance Band Violation

$$D_{\text{band}} = \sum_i \left[\text{sp}(\rho_i - \beta; \tau_{\text{out}}) + \text{sp}(\alpha - \rho_i; \tau_{\text{out}})\right]$$

6.2 Overdominance Indicator

$$O = \text{sp}(\max_i \rho_i - \theta; \tau_{\text{out}})$$

6.3 Rate of Change

$$G = \sqrt{\sum_i \dot{\rho}_i^2}$$

6.4 Internal Deviation

$$D_{\text{in}} = \sum_i d_i(p_i)$$

7 Hazard Function

Raw score:

$$h_{\text{raw}} = k_1 D_{\text{band}} + k_2 O + k_3 G + k_4 D_{\text{in}}$$

Normalized form:

$$\tilde{h} = 1 - \exp(-h_{\text{raw}}) \in [0, 1)$$

8 Phase Regimes

8.1 Phase A — Balanced TRIAS

$$\forall i : \alpha \leq \rho_i \leq \beta$$

$$\forall i : p_i \in [p^\star - \delta, p^\star + \delta]$$

8.2 Phase B — Drift

$$\neg\text{Balanced} \quad \wedge \quad \max_i \rho_i < \theta$$

8.3 Phase C — Overdominance

$$\rho_i \geq \theta \quad \wedge \quad \rho_i > \max_{j \neq i} \rho_j$$

9 Scale Invariance

$$u \mapsto cu, \quad c > 0$$

Leaves:

$$\rho_i, D_{\text{band}}, O, G, D_{\text{in}}, \tilde{h}, \text{Phase classification}$$

unchanged.
The model is structurally scale invariant.

10 Interpretation Summary (Engineering View)

- u_i: raw structural potency
- ρ_i: normalized domain share
- $d_i(p_i)$: internal 70/30 deviation penalty
- D_{band}: macro imbalance
- O: monopolization indicator
- G: instability velocity

- $\tilde{h}$: early warning hazard score